Random Forest Classification using Iris Dataset

1. Introduction

Machine learning is widely used to solve classification problems by identifying patterns in data and making predictions. In this project, the **Random Forest Classification algorithm** is applied to the **Iris Dataset** to classify iris flowers into different species based on their physical measurements.

The Iris dataset is one of the most popular and beginner-friendly datasets in machine learning. It contains flower measurements such as **sepal length**, **sepal width**, **petal length**, and **petal width**, which are used to classify flowers into one of three species:

- **Setosa**
- **Versicolor**
- **Virginica**

Random Forest is a supervised ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

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2. Aim

The aim of this project is to build a **Random Forest Classification model** using the **Iris Dataset** and evaluate its performance in classifying flower species accurately.

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3. Objectives

The objectives of this project are:

- To read the Iris dataset from a CSV file
- To understand the dataset structure
- To perform exploratory data analysis
- To visualize the class distribution
- To define features and target variable
- To split the dataset into training and testing sets
- To train a Random Forest Classifier
- To evaluate the model performance
- To analyze feature importance

4. Tools and Libraries Used

This project was implemented using **Python** and several important libraries for data handling, visualization, machine learning, and model evaluation.

The libraries used include:

- **pandas** for data manipulation
- **numpy** for numerical operations
- **matplotlib** and **seaborn** for visualization
- **scikit-learn** for model training and evaluation

These tools were used throughout the project for:

- reading and analyzing data
- plotting graphs
- training the Random Forest model
- making predictions
- evaluating classification performance

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5. Dataset Description

The **Iris Dataset** is a well-known classification dataset commonly used in machine learning.

It contains **150 rows** and **5 columns**.

Input Features

The four input features are:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable

The target variable is **species**, which contains three classes:

- **Setosa**
- **Versicolor**
- **Virginica**

Each class has **50 samples**, making the dataset **perfectly balanced**.

This dataset is ideal for classification because it is clean, small, and easy to understand.

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6. Data Exploration

The dataset was explored using:

- `describe()` to generate statistical summaries

This helped in understanding:

- the structure of the dataset
- the numerical range of the flower measurements
- whether there were any missing values

The Iris dataset is well-structured and does not contain missing values, which makes it suitable for direct analysis and model training.

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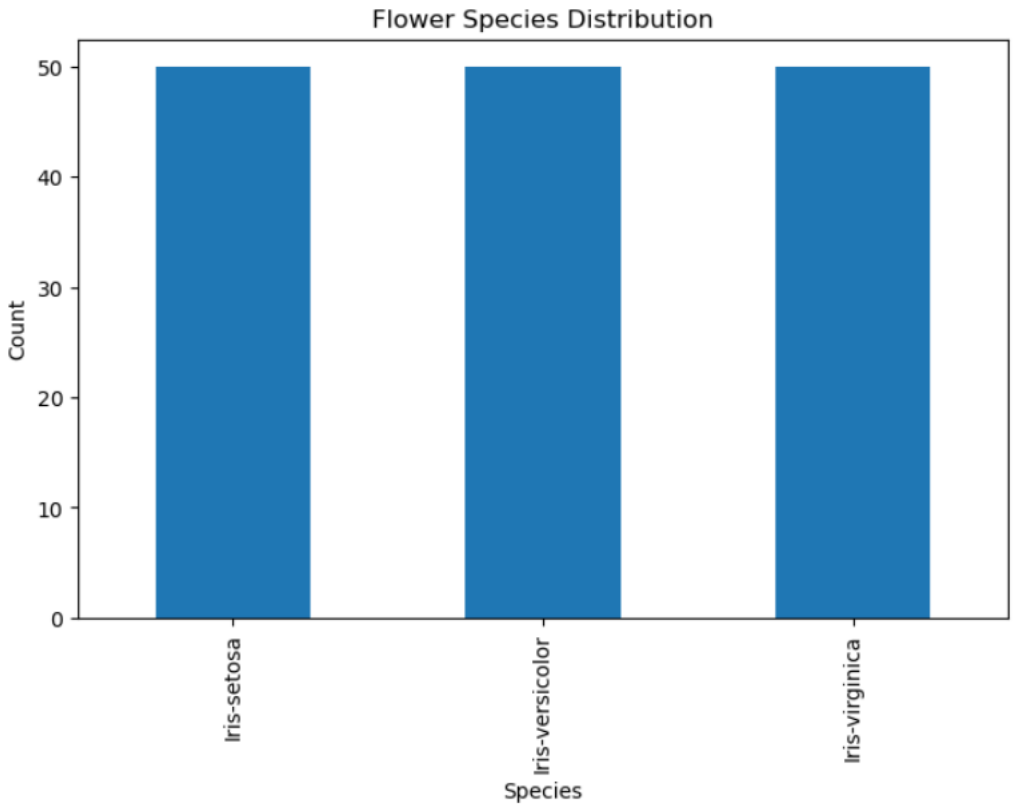
7. Class Distribution

The distribution of the flower species was checked to understand whether the dataset was balanced.

The dataset contains:

- **50 Setosa flowers**
- **50 Versicolor flowers**
- **50 Virginica flowers**

A **bar chart** was created to visually represent the class distribution. Since all three classes contain equal samples, the dataset is balanced and suitable for classification.



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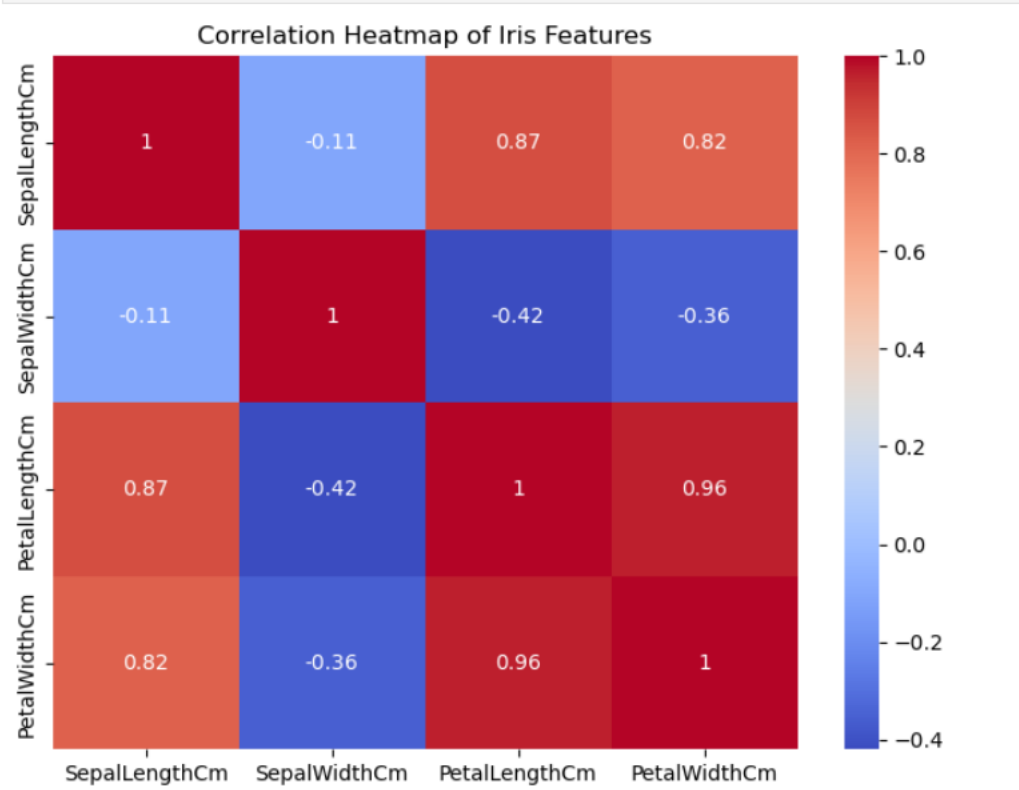
8. Correlation Analysis

A **correlation heatmap** was created to understand the relationship between the numerical features in the Iris dataset.

This helped in identifying:

- which flower measurements are strongly related
- which variables may contribute more to classification
- the relationship between petal and sepal dimensions

The heatmap provided useful insight into the dataset structure before training the model.



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9. Feature and Target Selection

The dataset was divided into:

- **X (Features)** → the input variables used for prediction
- **y (Target Variable)** → the flower species label

Here:

- **X** contains the four flower measurements
- **y** contains the flower species

This step is necessary before splitting the dataset and training the machine learning model.

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10. Splitting the Dataset

The dataset was split into:

- **80% Training Data**
- **20% Testing Data**

This split allows the Random Forest model to learn from one part of the data and then be tested on unseen samples.

This is important because it helps us measure how well the model generalizes to new data.

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11. Model Building

The machine learning algorithm used in this project is the **Random Forest Classifier**.

The model was initialized using:

- `n_estimators = 100`
- `random_state = 42`

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12. Prediction and Evaluation

After training the model, predictions were made on the testing dataset.

The model was evaluated using:

- **Accuracy Score**
- **Confusion Matrix**
- **Classification Report**

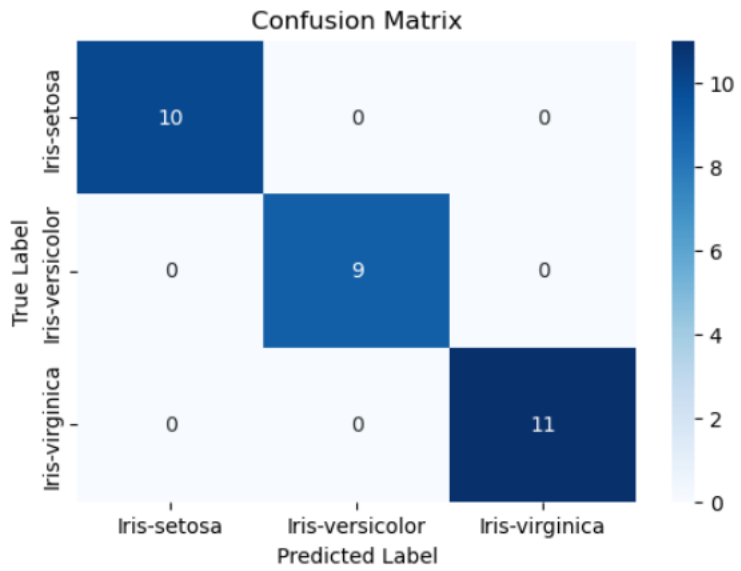
Accuracy

The Random Forest model achieved a very high accuracy on the Iris dataset, showing that it can classify flower species effectively.

Confusion Matrix

The confusion matrix helped in understanding:

- how many flowers were classified correctly
- whether any species were misclassified



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Classification Report

The classification report provided:

- **Precision**
- **Recall**
- **F1-score**
- **Support**

These metrics gave a more detailed understanding of model performance for each flower species.

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13. Feature Importance Analysis

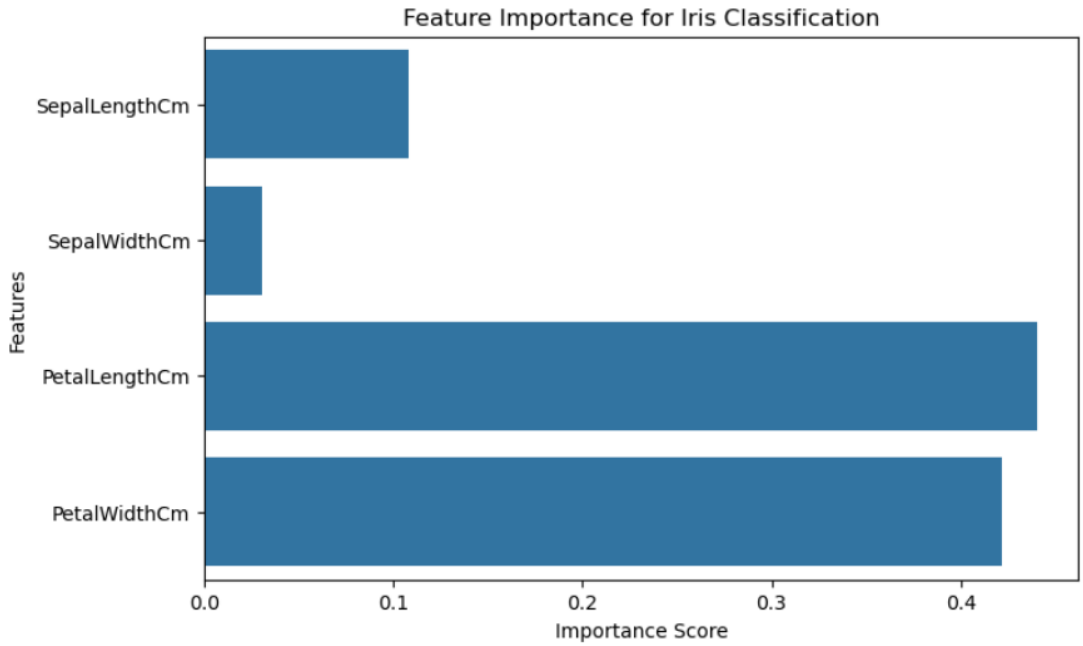
One of the advantages of Random Forest is that it can identify which input features contribute the most to prediction.

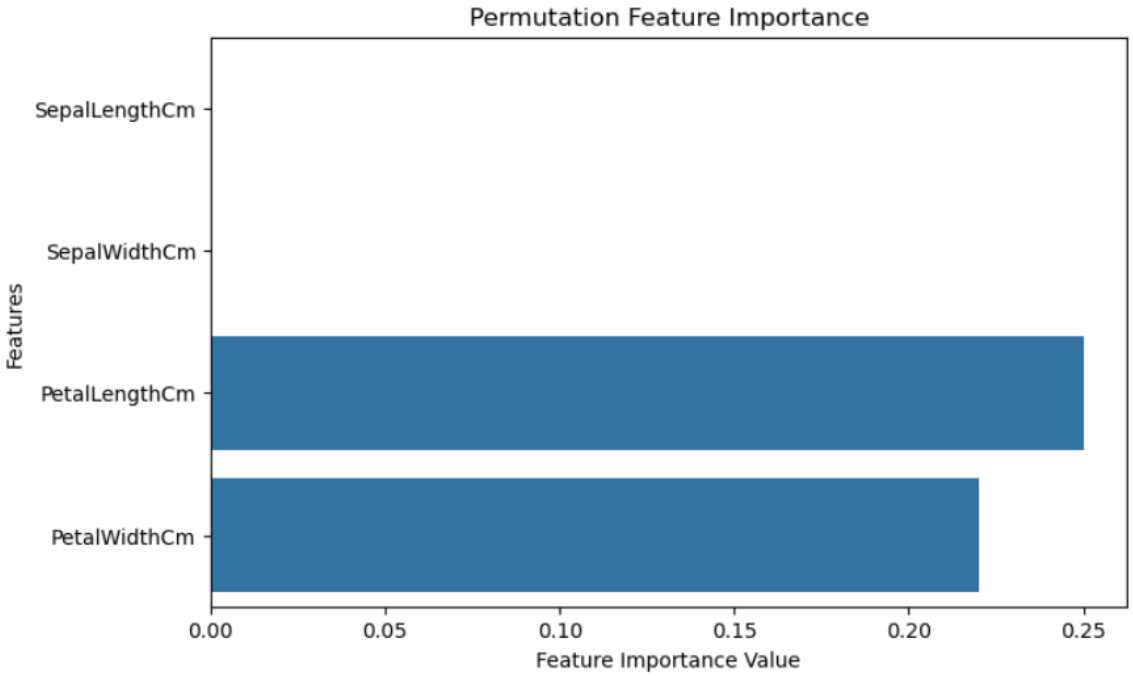
In this project, **feature importance analysis** was used to determine which flower measurements were most important in classifying the iris species.

This helps in understanding the significance of:

- sepal measurements
- petal measurements

This makes the model more interpretable and meaningful.





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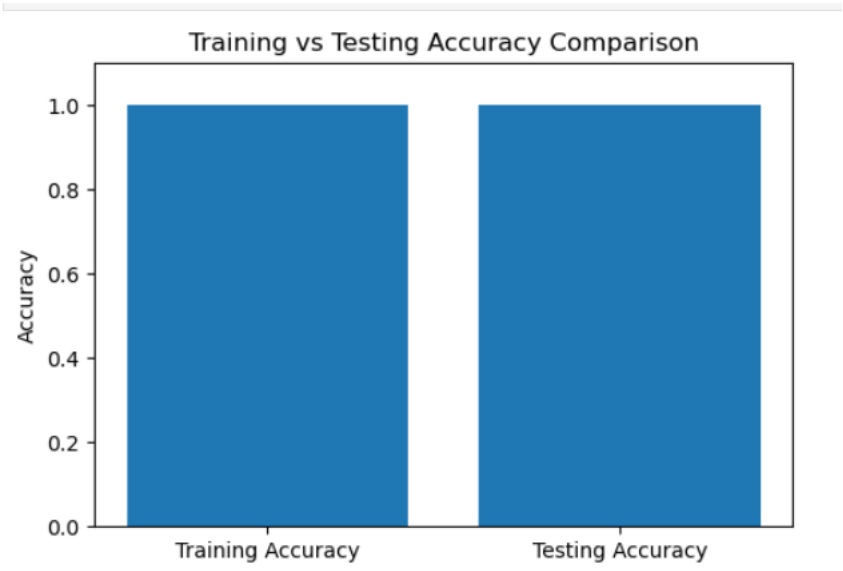
14. Accuracy Comparison

A comparison between **training accuracy** and **testing accuracy** was performed to check whether the model was:

- well-generalized
- overfitting
- underfitting

If both values are close, it usually means the model is performing consistently on both seen and unseen data.

An **accuracy comparison plot** was created to visually represent this comparison.



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16. Conclusion

In this project, the **Random Forest Classification algorithm** was successfully applied to the **Iris Dataset** to classify flowers into **Setosa**, **Versicolor**, and **Virginica**.

The dataset was explored, visualized, and prepared successfully. The Random Forest model was then trained and evaluated using standard classification metrics.

Final Findings

- The dataset was clean and balanced
- The model achieved high classification accuracy
- The evaluation metrics showed strong performance
- Feature importance helped interpret the model

Final Conclusion

This project demonstrates that **Random Forest** is a highly effective machine learning algorithm for **multi-class classification problems**.

It offers:

- high accuracy
- robustness
- interpretability
- good generalization ability

Overall, the project was successful in both **prediction performance** and **analytical understanding**.

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17. References

1. Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32.
2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer.
3. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
4. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
5. Fisher, R. A. (1936). *The use of multiple measurements in taxonomic problems*. Annals of Eugenics, 7(2), 179–188.

GITHUB LINK:

https://github.com/saiprasadduggirala27-source/Iris-Dataset-Individual-Project.git